

AI-BASED POOL MONITORING SYSTEM (NABEH)



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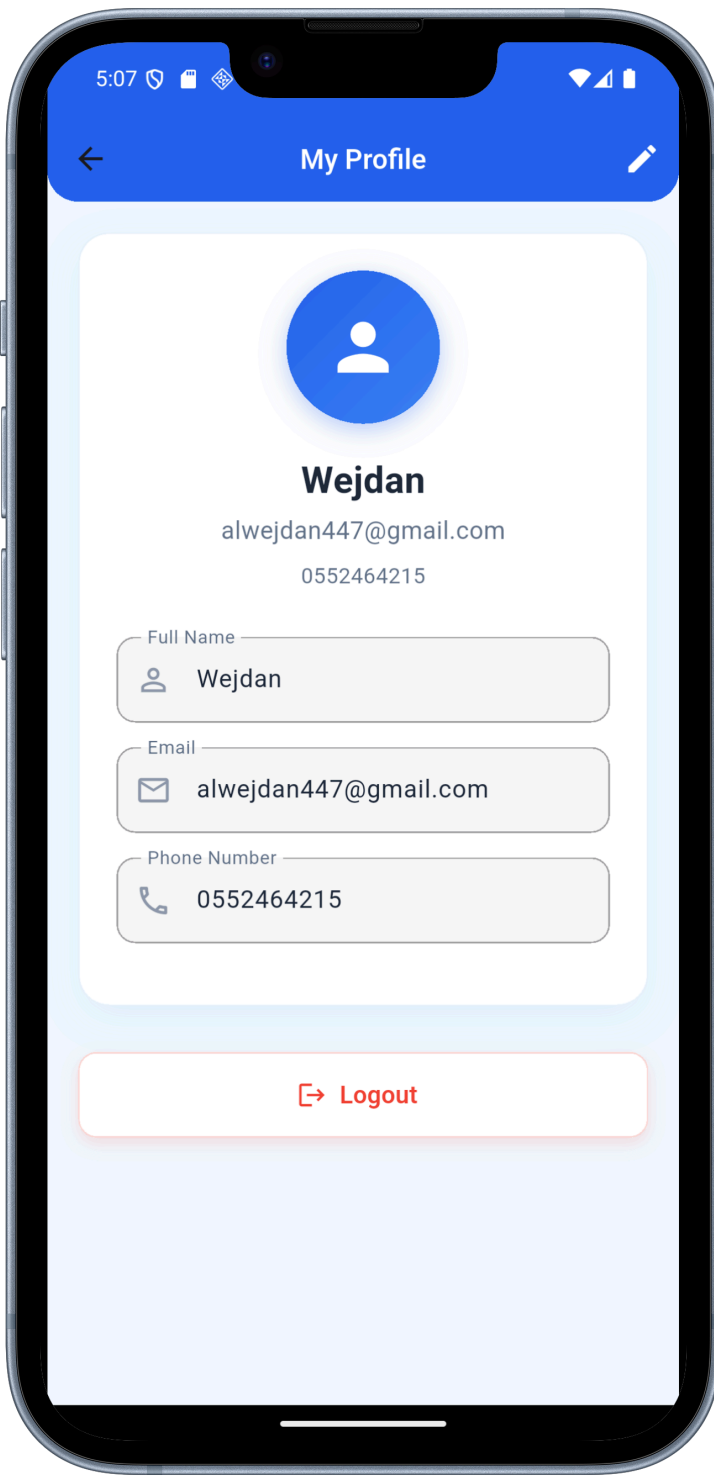
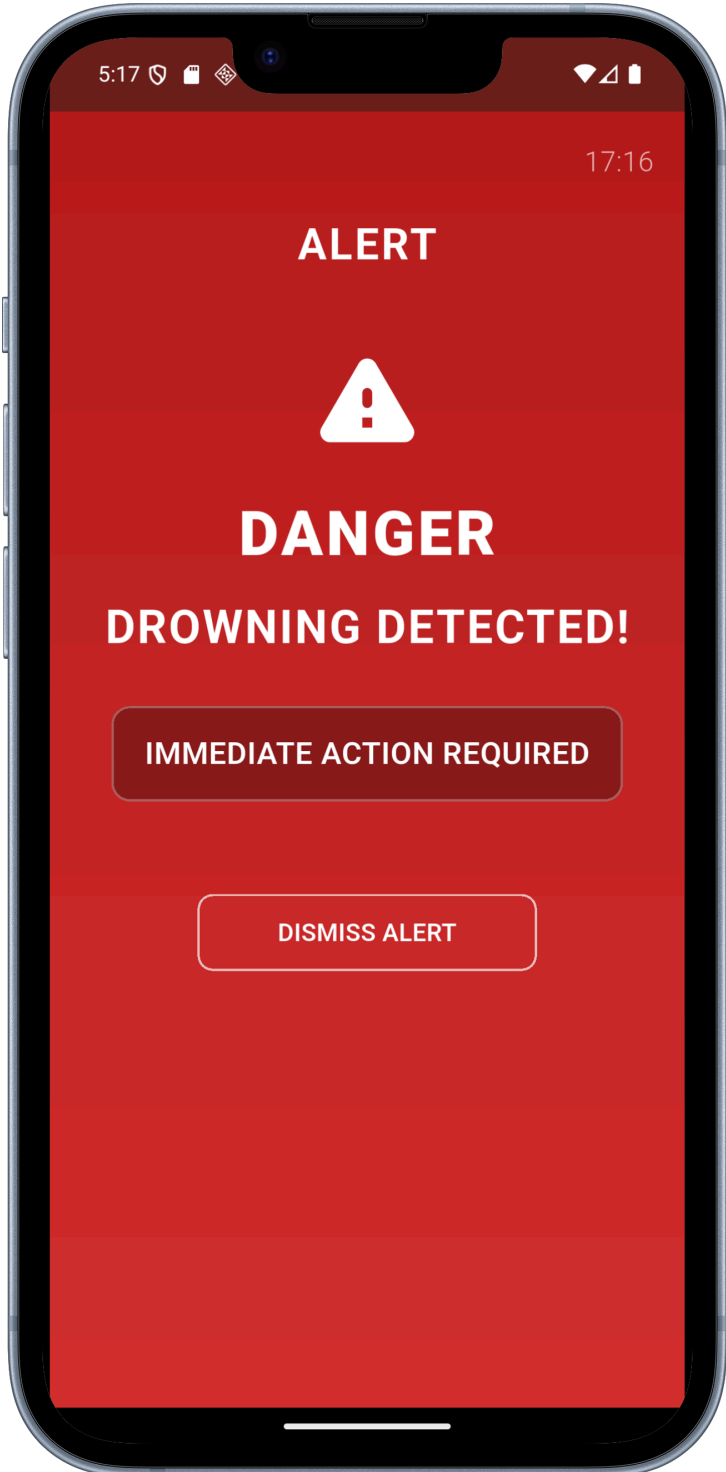
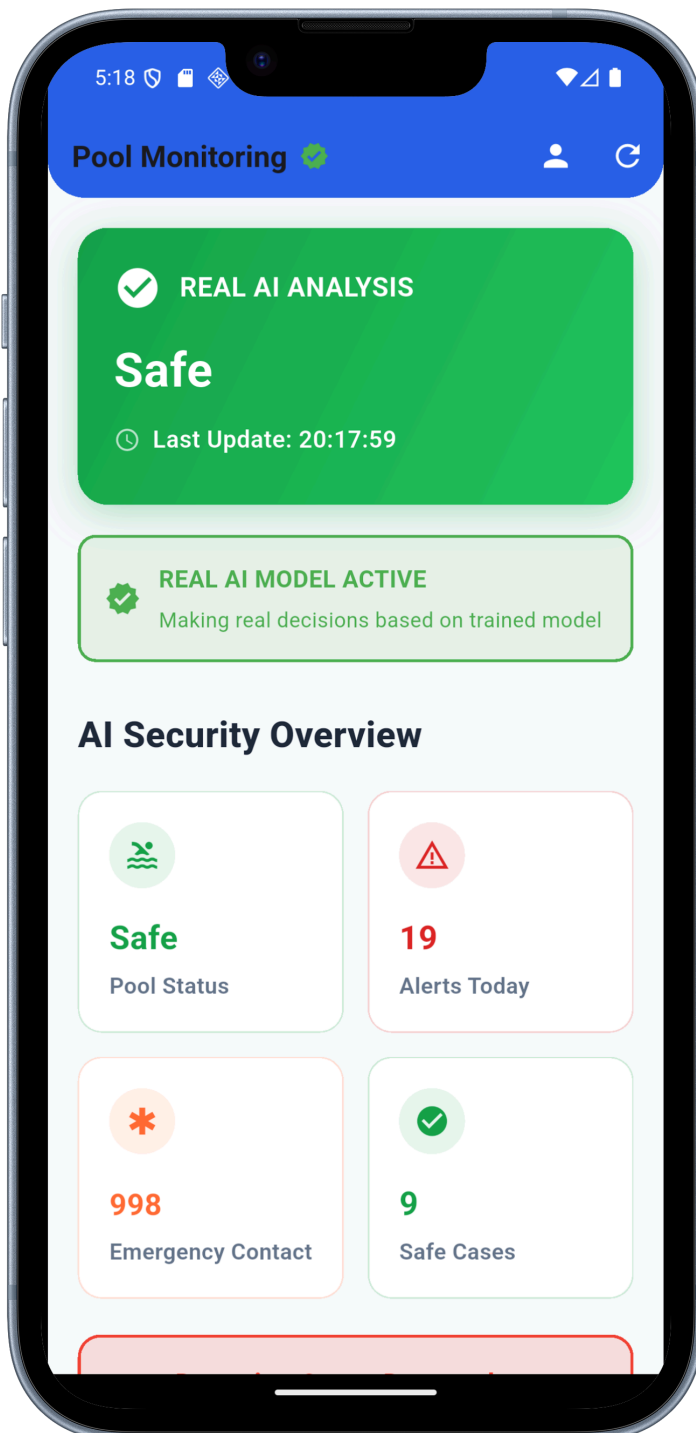
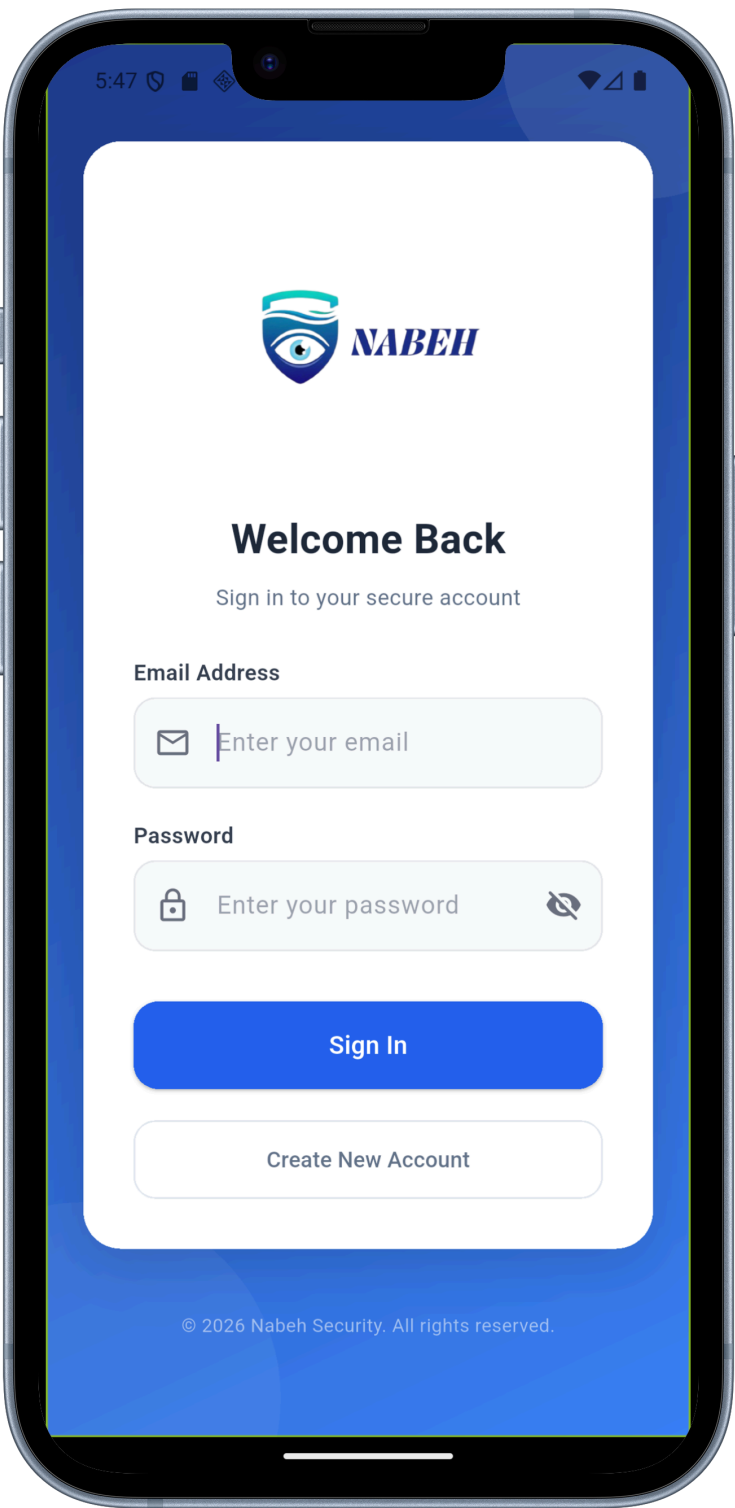
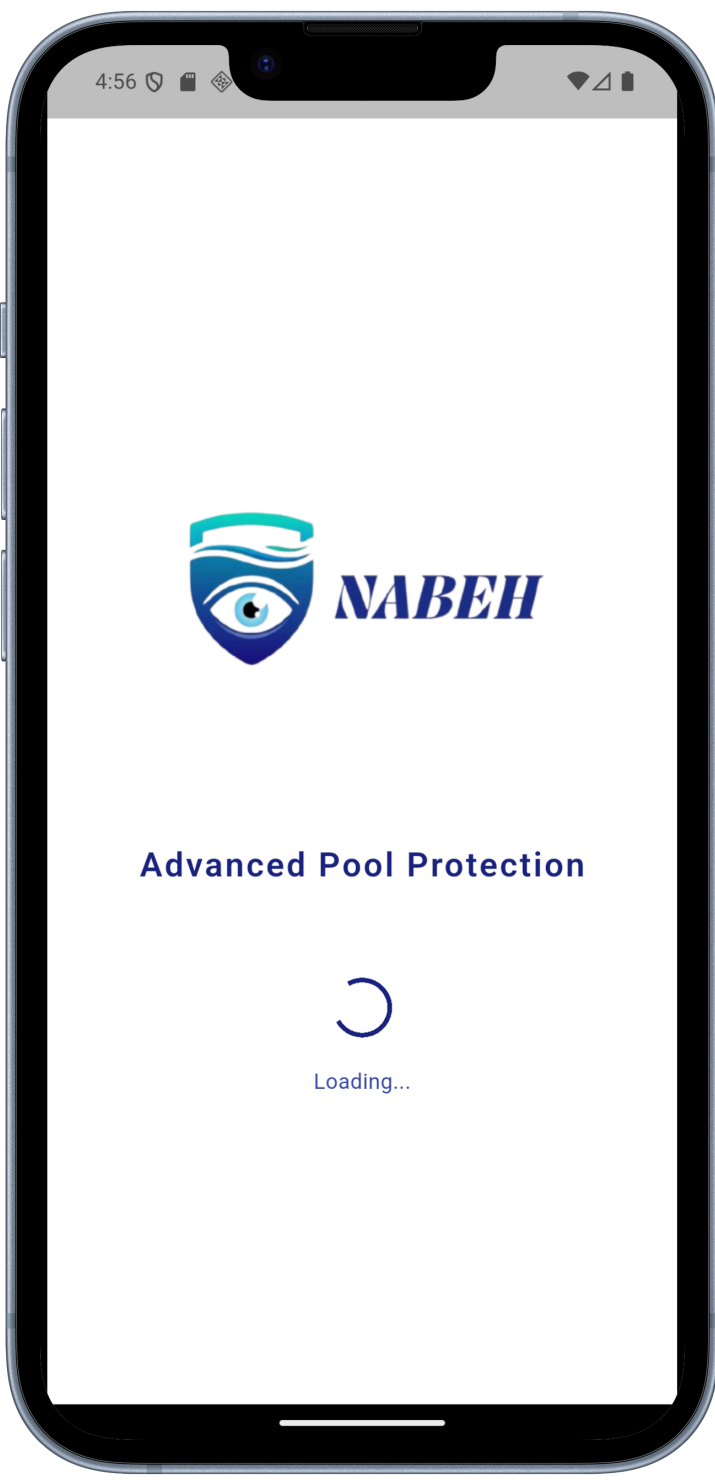
introduction

NEBEH is an AI-powered swimming pool monitoring system designed to detect potential drowning incidents in real time. The system analyzes swimmers’ behavior and sends instant alerts with audible notifications through a mobile application, enhancing safety and enabling faster emergency response.

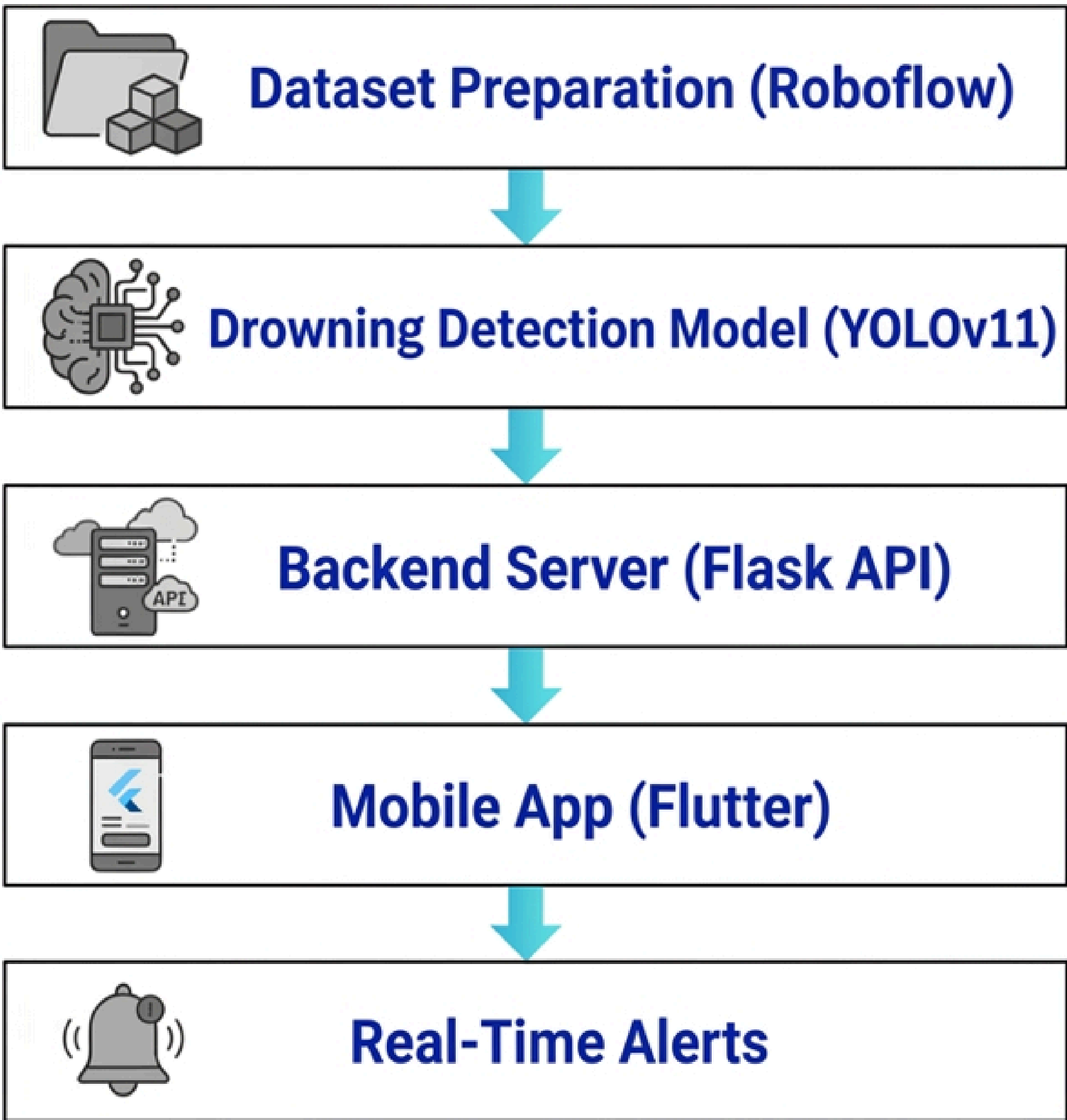
Experiments

Experiments were conducted using the YOLOv11 model to evaluate the NABEH system. A dataset of swimmer videos was used and split into training and testing sets. The model was trained to detect abnormal swimming movements indicating potential danger. Results showed high accuracy in identifying unusual behavior, demonstrating the system’s effectiveness for real-time safety monitoring.

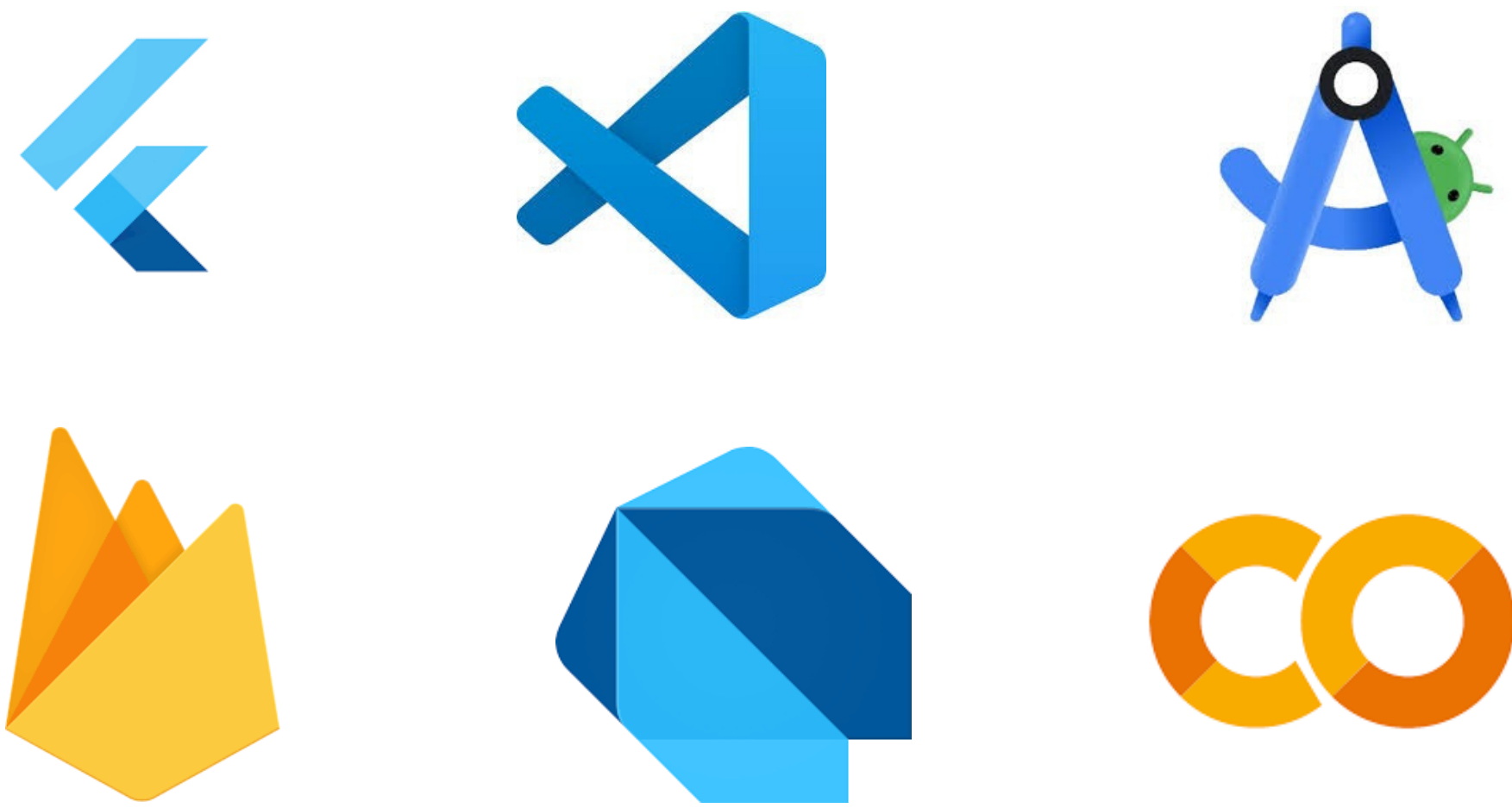
Result



Methodology



Tools



Conclusion

NEBEH proves the effectiveness of AI in improving pool safety through real-time monitoring and instant alerts, enabling faster response and reducing risks in aquatic environments.